Scalp EEG measurements are a staple in neuropsychiatric research and diagnostics but invariably suffer from poor spatial resolution and considerable contamination by noise, often limiting their immediate utility and demanding the use of invasive recording methods for applications such as seizure-focus identification, closed-loop deep-brain stimulation, or brain-machine interfaces. THAlis is a major step forward to increase the utility of scalp EEG recordings, in this case for the control of motor prostheses, by using manifold mapping and reservoir computing to increase decoding robustness over established methods. This progress is impressive compared to current standards in related academic fields, and absolutely outstanding when considering that it was implemented as a high-school research project.

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